

EXPERIMENT - 09

Aim: Relay-Based ON / OFF control system with Arduino.

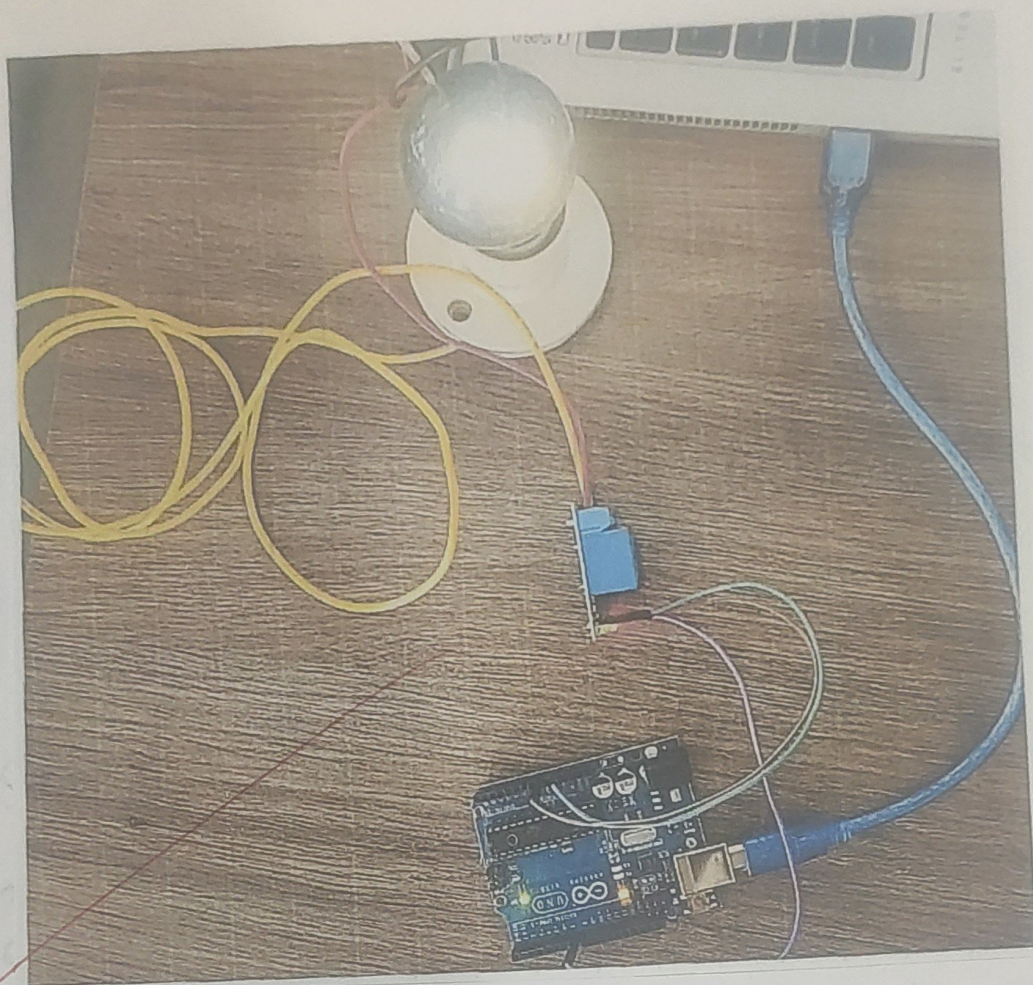
Appartus:

— Arduino UNO, Relay module, Bread board, wires, LED, bulb, Power supply, USB cable, Resistor.

Theory:

- DC- pin: needs to be connected to GND. (0V)
- DC+ pin: needs to be connected to VCC. (5V)
- IN pin: receives the control signal from Arduino.
- COM pin: is the common pin. It is used in both normally open mode and normally closed mode.
- NO pin: is normally open pin. It is used in the normally open mode.
- NC pin: is normally closed pin. It is used in the normally closed mode.

Performance result:



Conclusion:

The Raspberry Pi 4 is a powerful device that can be used for many different purposes.

It is a great device for learning about computers and electronics.

It is also a great device for creating projects.

With a little bit of creativity, you can do so much with it.

It is a great device for anyone who is interested in technology.

Experiment - 01

Annexure No.:

CODE:

```
int relayPin = 7;
void setup() {
    pinMode (relayPin, OUTPUT);
}
void loop() {
    digitalWrite (relayPin, HIGH);
    delay (1000);
    digitalWrite (relayPin, LOW);
    delay (1000);
}
```

Conclusion:

A relay-based control system using Arduino Uno enables safe and efficient ON/OFF control of high-power devices through low-power signals, making it useful for automation and control applications.

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